

## SANYA HEDONG, China

WMO: 599485

Lat: 18.2508N

Lon: 109.5077E

Elev: 23

StdP: 14.68

Time Zone: 8.00 (E08)

Period: 94-08

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | 61.3       | 63.3 | 45.3                          | 44.8 | 67.2 | 49.7 | 52.9 | 66.9 | 10.2                  | 74.5 | 9.5 | 74.4 | 3.2                   | 40   | 0.282 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 6             | 8.0                    | 91.7            | 79.9 | 90.8 | 79.9 | 90.0 | 79.8 | 82.6                | 88.4 | 81.9 | 87.8 | 81.6 | 87.4 | 5.1                  | 250  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 80.9                            | 160.9 | 86.5 | 80.4 | 158.0 | 86.1 | 79.9 | 155.4 | 85.8 | 46.5          | 88.7 | 46.0 | 88.0 | 45.4 | 87.7 | 84.4              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| 10.8              | 9.2  | 7.8 | 56.7                       | 93.9 | 3.4                | 1.4 | 54.3                                               | 94.9 | 52.3       | 95.8 | 50.4       | 96.6 | 47.9       | 97.6 |    |    |
|                   |      |     | 50.1                       | 83.3 | 3.9                | 0.6 | 47.3                                               | 83.8 | 45.0       | 84.1 | 42.8       | 84.5 | 39.9       | 84.9 |    |    |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 80.4  | 73.2  | 74.4  | 78.2  | 82.4  | 84.3  | 85.2  | 84.5  | 84.0  | 83.0  | 81.9  | 78.5  | 74.6 | (6)  |
| (7)  |                                                                                  | DBStd       | 5.06  | 3.39  | 3.86  | 3.59  | 2.39  | 2.30  | 2.18  | 2.14  | 1.83  | 2.03  | 2.22  | 3.13  | 4.07 | (7)  |
| (8)  |                                                                                  | HDD50       | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | (8)  |
| (9)  |                                                                                  | HDD65       | 4     | 1     | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 2    | (9)  |
| (10) |                                                                                  | CDD50       | 11085 | 719   | 683   | 873   | 973   | 1062  | 1057  | 1069  | 1054  | 989   | 988   | 856   | 762  | (10) |
| (11) |                                                                                  | CDD65       | 5615  | 255   | 264   | 408   | 523   | 597   | 607   | 604   | 589   | 539   | 523   | 406   | 300  | (11) |
| (12) |                                                                                  | CDH74       | 57242 | 1001  | 1257  | 3101  | 5723  | 7415  | 7916  | 7589  | 7193  | 6063  | 5362  | 3169  | 1453 | (12) |
| (13) |                                                                                  | CDH80       | 20313 | 80    | 150   | 648   | 1967  | 3147  | 3665  | 3233  | 2843  | 2081  | 1619  | 695   | 185  | (13) |
| (14) | Wind                                                                             | WSAvg       | 3.6   | 3.4   | 3.4   | 3.4   | 3.3   | 3.4   | 3.5   | 3.6   | 3.7   | 3.6   | 3.7   | 3.8   | 3.8  | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 49.1  | 0.2   | 0.2   | 0.4   | 1.3   | 4.3   | 7.4   | 6.0   | 8.0   | 10.2  | 8.9   | 1.7   | 0.3  | (15) |
| (16) |                                                                                  | PrecMax     | 78.3  | 1.4   | 1.1   | 2.0   | 3.9   | 17.6  | 16.3  | 10.6  | 19.2  | 24.8  | 21.9  | 11.8  | 1.2  | (16) |
| (17) |                                                                                  | PrecMin     | 26.5  | 0.0   | 0.0   | 0.0   | 0.0   | 0.4   | 0.2   | 0.7   | 1.5   | 1.9   | 0.5   | 0.0   | 0.0  | (17) |
| (18) |                                                                                  | PrecStd     | 13.5  | 0.3   | 0.2   | 0.6   | 1.1   | 3.9   | 3.6   | 3.0   | 3.3   | 5.3   | 6.0   | 2.3   | 0.3  | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 83.7  | 84.8  | 88.1  | 90.9  | 92.6  | 93.3  | 91.8  | 91.7  | 91.8  | 90.7  | 89.0  | 86.1 | (19) |
| (20) |                                                                                  |             | MCWB  | 72.2  | 72.8  | 75.6  | 77.8  | 79.8  | 80.5  | 80.4  | 80.2  | 78.5  | 76.9  | 76.3  | 74.6 | (20) |
| (21) |                                                                                  | 2%          | DB    | 82.1  | 83.3  | 86.2  | 89.6  | 91.2  | 91.7  | 90.7  | 90.2  | 90.1  | 89.2  | 87.1  | 83.8 | (21) |
| (22) |                                                                                  |             | MCWB  | 71.4  | 72.7  | 75.5  | 78.0  | 79.9  | 80.9  | 80.4  | 80.3  | 78.5  | 76.7  | 75.0  | 72.8 | (22) |
| (23) |                                                                                  | 5%          | DB    | 80.7  | 82.0  | 84.9  | 88.5  | 90.2  | 90.6  | 89.9  | 89.3  | 88.9  | 88.1  | 85.6  | 82.0 | (23) |
| (24) |                                                                                  |             | MCWB  | 70.6  | 72.1  | 75.0  | 77.8  | 79.8  | 80.9  | 80.4  | 80.2  | 78.5  | 76.6  | 74.1  | 71.4 | (24) |
| (25) |                                                                                  | 10%         | DB    | 79.3  | 80.6  | 83.7  | 87.4  | 89.1  | 89.5  | 89.0  | 88.4  | 87.8  | 86.9  | 84.0  | 80.4 | (25) |
| (26) |                                                                                  |             | MCWB  | 69.7  | 71.3  | 74.5  | 77.6  | 79.7  | 80.8  | 80.4  | 80.1  | 78.5  | 76.2  | 73.3  | 70.4 | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 75.1  | 76.4  | 77.8  | 81.0  | 82.7  | 83.6  | 82.8  | 82.7  | 81.6  | 80.0  | 78.2  | 76.6 | (27) |
| (28) |                                                                                  |             | MCDWB | 79.9  | 81.2  | 84.1  | 86.9  | 88.9  | 89.5  | 88.5  | 88.2  | 87.5  | 87.1  | 85.2  | 82.9 | (28) |
| (29) |                                                                                  | 2%          | WB    | 73.1  | 75.0  | 77.0  | 80.1  | 81.8  | 82.7  | 82.2  | 81.9  | 80.7  | 79.1  | 77.3  | 75.2 | (29) |
| (30) |                                                                                  |             | MCDWB | 80.1  | 80.3  | 83.4  | 86.1  | 88.1  | 88.6  | 87.9  | 87.4  | 86.7  | 85.9  | 83.7  | 80.3 | (30) |
| (31) |                                                                                  | 5%          | WB    | 72.0  | 73.7  | 76.3  | 79.4  | 81.2  | 82.1  | 81.7  | 81.4  | 80.0  | 78.4  | 76.4  | 74.0 | (31) |
| (32) |                                                                                  |             | MCDWB | 78.5  | 79.5  | 82.6  | 85.6  | 87.4  | 88.0  | 87.4  | 86.9  | 86.0  | 84.9  | 82.6  | 79.6 | (32) |
| (33) |                                                                                  | 10%         | WB    | 70.9  | 72.7  | 75.6  | 78.7  | 80.7  | 81.6  | 81.1  | 80.8  | 79.4  | 77.8  | 75.6  | 72.2 | (33) |
| (34) |                                                                                  |             | MCDWB | 77.3  | 78.6  | 81.8  | 85.1  | 86.8  | 87.5  | 86.9  | 86.4  | 85.3  | 83.9  | 81.8  | 78.8 | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR  | 11.2  | 10.2  | 9.5   | 9.2   | 8.8   | 8.0   | 7.7   | 8.0   | 9.0   | 10.1  | 10.7  | 10.6 | (35) |
| (36) |                                                                                  |             | MCDBR | 12.1  | 11.5  | 10.2  | 10.1  | 9.8   | 8.8   | 8.7   | 9.5   | 11.1  | 11.6  | 12.3  | 11.7 | (36) |
| (37) |                                                                                  |             | MCWBR | 4.9   | 4.2   | 3.5   | 3.1   | 3.3   | 3.2   | 3.1   | 3.5   | 4.0   | 4.4   | 4.7   | 5.1  | (37) |
| (38) |                                                                                  | 5% WB       | MCDWB | 10.9  | 9.2   | 8.7   | 8.5   | 8.7   | 7.6   | 7.7   | 8.3   | 8.8   | 9.5   | 9.6   | 10.1 | (38) |
| (39) |                                                                                  |             | MCWBR | 4.9   | 3.9   | 3.3   | 3.2   | 3.5   | 3.3   | 3.4   | 3.8   | 3.9   | 4.0   | 3.9   | 4.4  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.507 | 0.525 | 0.627 | 0.591 | 0.483 | 0.470 | 0.451 | 0.475 | 0.531 | 0.570 | 0.514 | 0.533 | (40) |      |
| (41) |                                                                                  | taud        | 2.022 | 2.013 | 1.791 | 1.932 | 2.285 | 2.336 | 2.387 | 2.302 | 2.106 | 1.946 | 2.074 | 1.961 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 244   | 248   | 228   | 237   | 261   | 262   | 267   | 263   | 248   | 233   | 241   | 231   | (42) |      |
| (43) |                                                                                  | Edh at Noon | 52    | 55    | 71    | 62    | 43    | 41    | 39    | 42    | 51    | 58    | 49    | 54    | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 1171  | 1367  | 1487  | 1741  | 1834  | 1721  | 1695  | 1613  | 1517  | 1424  | 1278  | 1102  | (44) |      |
| (45) |                                                                                  | RadStd      | 145   | 189   | 110   | 135   | 123   | 142   | 163   | 115   | 157   | 172   | 106   | 112   | (45) |      |

## Historical Trends

|      |                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|      |                        | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) | Station Only           | N/A   | N/A     | N/A    | +1.28   | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) | Regional (5 neighbors) | +0.47 | N/A     | N/A    | +0.58   | N/A   | N/A   | N/A         | N/A   | +162  | +149  |

Nomenclature: See separate page